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Gut microbiota profiling in aged dogs after feeding pet food contained *Hericium erinaceus*

Hyun-Woo Cho[#], Soyoung Choi[#], Kangmin Seo, Ki Hyun Kim, Jung-Hwan Jeon, Chan Ho Kim, Sejin Lim, Sohee Jeong and Ju Lan Chun*

National Institute of Animal Science, Rural Development Administration, Wanju 55365, Korea

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[#]These authors contributed equally to this work.

***Corresponding author**
Ju Lan Chun
National Institute of Animal Science, Rural Development Administration, Wanju 55365, Korea.
Tel: +82-63-238-7053
E-mail: julanchun@korea.kr

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ORCID
Hyun-Woo Cho
https://orcid.org/0000-0002-3620-9952
Soyoung Choi
https://orcid.org/0000-0003-0045-3758
Kangmin Seo
https://orcid.org/0000-0001-6152-8536
Ki Hyun Kim
https://orcid.org/0000-0002-9834-2126
Jung-Hwan Jeon
https://orcid.org/0000-0001-9725-547X
Chan Ho Kim
https://orcid.org/0000-0003-2121-5249
Sejin Lim
https://orcid.org/0000-0002-0465-1666
Sohee Jeong
https://orcid.org/0000-0001-8325-162X
Ju Lan Chun
https://orcid.org/0000-0002-4618-586X

Abstract

Health concern of dogs is the most important issue for pet owners. People who have companied the dogs long-term provide the utmost cares for their well-being and healthy life. Recently, it was revealed that the population and types of gut microbiota affect the metabolism and immunity of the host. However, there is little information on the gut microbiome of dogs. *Hericium erinaceus* (*H. erinaceus*; HE) is one of the well-known medicinal mushrooms and has multiple bioactive components including polyphenol, β-glucan, polysaccharides, ergothioneine, hericerin, erinacines, etc. Here we tested a pet food that contained *H. erinaceus* for improvement in the gut microbiota environment of aged dogs. A total of 18 dogs, each 11 years old, were utilized. For sixteen weeks, the dogs were fed with 0.4 g of H. erinaceus (HE-L), or 0.8 g (HE-H), or without *H. erinaceus* (CON) per body weight (kg) with daily diets (n = 6 per group). Taxonomic analysis was performed using metagenomics to investigate the difference in the gut microbiome. Resulting from principal coordinates analysis (PCoA) to confirm the distance difference between the groups, there was a significant difference between HE-H and CON due to weighted Unique fraction metric (Unifrac) distance (p = 0.047), but HE-L did not have a statistical difference compared to that of CON. Additionally, the result of Linear discriminate analysis of effect size (LEfSe) showed that phylum *Bacteroidetes* in HE-H and its order Bacteroidales increased, compared to that of CON, Additionally, phylum *Firmicutes* in HE-H, and its genera (*Streptococcus*, *Tyzzerella*) were reduced. Furthermore, at the family level, *Campylobacteraceae* and its genus *Campylobacter* in HE-H was decreased compared to that of CON. Summarily, our data demonstrated that the intake of *H. erinaceus* can regulate the gut microbial community in aged dogs, and an adequate supply of HE on pet diets would possibly improve immunity and anti-obesity on gut-microbiota in dogs.

Keywords: *Hericium erinaceus*, Aged dogs, Gut microbiota, Metagenomic analysis, Taxonomic profile

INTRODUCTION

The population of dogs as companion animals has increased in the last few decades. The need to provide proper cares for the health and well-being of dogs has also increased, such as premium pet foods and advanced medical services. As dogs receive more care, their health and lifespan expand, which increases

Competing interests
No potential conflict of interest relevant to this article was reported.

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Availability of data and material
Upon reasonable request, the datasets of this study can be available from the corresponding author.

Authors' contributions
Conceptualization: Cho HW, Choi S, Chun JL.
Data curation: Cho HW, Choi S, Seo K.
Formal analysis: Cho HW, Choi S, Seo K, Kim KH.
Methodology: Kim KH, Kim CH, Lim S, Jeong S.
Validation: Seo K, Kim KH, Jeon JH.
Writing – original draft: Cho HW, Chun JL.
Writing –review & editing: Cho HW, Choi S, Seo K, Kim KH, Jeon JH, Kim CH, Lim S, Jeong S, Chun JL.

Ethics approval and consent to participate
Animal studies were approved by the Institutional Animal Care and Use Committee of the National Institute of Animal Science (NIAS), Korea (Approval number: NIAS-2020-438).

the geriatric population of dogs [1]. Like humans, dogs experience age-associated changes in a variety of physiological and metabolic aspects [2]. To maintain homeostasis in body systems it is important to provide essential and balanced nutrients. Currently, it is also common to feed dogs pet foods with functional ingredients which are used to help human health to delay aging or prevent the development of age-related diseases [3–5]. However, the current knowledge on proper nutrients and ingredients for aged dogs is limited.

Hericium erinaceus (*H. erinaceus*; HE) also known as Lion's mane mushroom is a widely used medicinal mushroom with an extensive history. *H. erinaceus* contains several biologically active substances which are responsible for anti-cancer, anti-inflammation, anti-oxidation, etc. [6–10]. The most well-known medicinal substances of *H. erinaceus* are erinacines and hericerin which stimulate nerve growth factor (NGF) biosynthesis [11]. NGF acts as a neurotrophic factor and is involved in the survival, growth, and maintenance of neuronal cells [12–14]. Ergothioneine is a naturally occurring betaine amino acid produced by only certain bacteria and fungi [15]. Mushrooms are rich sources of ergothioneine including *H. erinaceus* [16]. The amount of ergothioneine in various mushrooms is correlated with the antioxidant capacity measured by *in-vitro* tests [17], and it was demonstrated that ergothioneine acts as an antioxidant *in vivo* and *in vitro* [18]. *H. erinaceus* also contains a large number of polyphenols that act as strong antioxidants and, anti-inflammatory substances and have anti-cancerous roles [19]. *H. erinaceus* extract administration prevented neurons from ischemic injury and improved cognitive function [20,21]. Additionally, *H. erinaceus* treatment in vascular endothelial cells induced antioxidant activity with antiangiogenic and anti-inflammatory activities, which may have anti-cancer properties [22]. Although *H. erinaceus* has various beneficial aspects, there are few studies on its effects on the health of dogs.

The importance of the gut microbiome in host health was widely researched. Gut microbiota digests variable prebiotics of non-digestible carbohydrates and produces fermentative by-products that can contribute to the health of the host [23]. *H. erinaceus* can be used as prebiotics for gut microbiota that could transforms polyphenols in *H. erinaceus* into a more bioactive low-molecular weight metabolite [24]. Additionally, the supplementation of *H. erinaceus* increased gut microbiota producing short-chain fatty acids in healthy adults that can be used by intestinal epithelial cells to enhance the gut barrier function [25]. Furthermore, it is reported that *H. erinaceus* could influence the relative abundance of beneficial gut microbiota composition, such as *Lachnospiraceae* and *Akkermansiaceae* [26]. It is reported that the specific gut microbiome pattern can be related to aging which could predict survival in humans [27]. Besides the accumulated information on *H. erinaceus* in gut microbiota, their effects on the gut microbiota in aged dogs have not been investigated. Hence, we evaluated the influence of *H. erinaceus* on the gut microbiome of dogs by taxonomic profiling using metagenomics. The resulting finding would be an essential source to study the gut microbiota, which can be used to provide better health care for aged dogs.

MATERIALS AND METHODS

Animals

Animal studies were approved by the Institutional Animal Care and Use Committee of the National Institute of Animal Science (NIAS), Korea (Approval number: NIAS-2020-438). Each dogs was housed individually indoors. The housing environment was maintained with constant temperature (22℃–24℃) and humidity (60%–80%). Dogs were fed once a day and water were provided *ad libitum*. All dogs were exercised outside in the playground once a day. The health of dogs was looked after and monitored daily by NIAS veterinarians in need.

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Diets

In this study, 5 Schnauzers, 6 Poodles, and 7 Maltese dogs were used. All the dogs were females and neutralized. The dogs were randomly distributed into three groups based on their similar body condition scores and body weights . All dogs were fed the same commercial diet except *H. erianceus*. A total of 6 dogs (aged 11 years) were fed diets without *H. erinaceus*, 6 dogs were fed 0.4 g per body weight (kg), and 6 dogs were fed 0.8 g per body weight (kg) of *H. erinaceus* for 16 weeks. The ingredient composition of the experimental diet is represented in Table 1. The amount of daily food intake was determined according to the recommendation of AFFCO (metabolizable energy [ME], kcal/day 132 × body weight 0.75 kg).

Sample preparation and 16S rRNA sequencing

The feces were collected freshly and frozen in liquid nitrogen immediately. Collected feces were delivered to the laboratory and stored at −80℃ until used. Genomic DNA (gDNA) was extracted from the frozen feces with the QIAmp PowerFecal DNA kit (QIAGEN, Hilden, Germany) according to the manufacturer's instructions. After quality control, the qualified samples proceeded to library construction. gDNA was amplified by V3-V4 (341F/805R) region specific primers (341F: 5′-CCTACGGGNGGCWGCAG-3′, 805R: 5′- GACTACHVGGGTATCTAATCC-3′). For Next Generation Sequencing (NGS), index sequences were added using Illumina Nextera XT Sample Preparation Kit (Illumina, San Diego, CA, USA). The Fastq files of 16S rRNA sequencing data were produced in the read lengths of 2 × 300 paired-end using the Illumina MiSeq platform (Illumina).

Microbiota analysis

The Fastq files were confirmed by Quantitative Insights Into Microbial Ecology (QIIME2 ver. 2020.11) [28]. First, primer sequences on raw sequencing were trimmed by the Cutadapt plugin [29]. Then, to obtain the amplicon sequencing variants (ASVs), the DADA2 plugin [30] was used to denoise under Q score 25 and chimeric sequence. For taxonomy classification, the Silva full-length (ver. SSU138) reference sequences were pre-trained V3-V4 primer by naive-bayes classifier

Table 1. Ingredients of experimental diets without or with *Hericium erinaceus*

Ingredients (%)	CON	HE - L	HE - H
Hericium erinaceus powder	-	1.24	2.48
Rice powder	31.9	30.7	29.4
Chicken breast powder	15.0	15.0	15.0
Yolk powder	12.0	12.0	12.0
Lard	1.5	1.5	1.5
Green laver	1.0	1.0	1.0
Calcium carbonate	1.0	1.0	1.0
Cabbage powder	1.0	1.0	1.0
Potassium citrate	0.6	0.6	0.6
Calcium phosphate	0.4	0.4	0.4
Vitamin and mineral	0.4	0.4	0.4
Salt	0.2	0.2	0.2
Water	35.0	35.0	35.0
Total	100	100	100

CON; group fed with a diet without *Hericium erinaceus*, HE-L; group fed with a diet of 0.4 g *Hericium erinaceus*, HE-H; group fed with a diet of 0.8 g *Hericium erinaceus*; n = 6 per group.

[31]. After assigning taxonomy, metrics of alpha and beta diversities were examined with respect to Shannon index, Chao1, Observed features, Evenness, and weighted and unweighted unique fraction metric (UniFrac) distance, respectively. The significance of alpha diversity was calculated using the Kruskal-Wallis (pairwise) in the QIIME2 tool. Statistical results of Principal coordinate analysis (PCoA) were completed using Permutational multivariate analysis of variance (PERMANOVA). To investigate the effect of *H. erinaceus* on the difference in gut microbiota, the Linear discriminant analysis (LDA) effect size (LEfSe) method was used LEfSe tool [32]. LEfSe was conducted by Kruskal-Wallis sum-rank test and the threshold for the LDA score was over 3.0. Statistical analyses of LEfSe were completed using the LEfSe plugin.

RESULTS AND DISCUSSION

DNA sequencing throughput stats and microbiome diversity

The Phred quality scores of raw data produced over 33, and the amount of average reads was 102,166 ± 7,215 in eighteen samples (Table 2). The low-quality reads were filtered (quality score < 25) after importing data at QIIME2 (ver. 2020.11), trimmed using primers for sequencing, and then chimera reads were removed according to consensus options. The average number of ASVs for microbiome analysis was 60,321 ± 5,272. Alpha diversity of species evenness (Evenness estimator), richness (Observed features, Chao1 index), and diversity (Shannon index) were analyzed to estimate the diversity of microbiota in each sample (Table 3). In the control group (CON) fed with a diet without HE, the observed features were 192.4 ± 39.22, Shannon index was 5.6 ± 0.37, evenness was 0.7 ± 0.04, and Chao1 index was 192.6 ± 39.15. In the group fed with a diet of 0.4

Table 2. Result of DADA2 statistics of 16s rRNA amplicon sequencing variants on the gut-microbiota in dogs after feeding diets with *Hericium erinaceus*

Sample-id	Group	Input	Filtered	Denoise	Merged	Non-chimeric
C2-1	CON	92526	80625	79394	74900	53305
C2-2	CON	106559	93358	92727	83277	58403
C2-3	CON	108517	95954	94368	89609	67646
C2-4	CON	109942	96474	93438	86881	65839
C2-5	CON	107238	96234	94123	87696	64505
C2-6	CON	111296	98599	96432	89855	66081
T1-1	HE-L	112033	98977	97648	92921	59231
T1-2	HE-L	97660	86009	83711	77688	54637
T1-3	HE-L	96495	84262	82119	76035	56370
T1-4	HE-L	106150	93839	91346	84423	63920
T1-5	HE-L	105953	93036	91595	87238	64976
T1-6	HE-L	106342	93807	91513	85083	61415
T2-1	HE-H	101852	90557	89064	84894	64756
T2-2	HE-H	98872	86882	84837	78323	56512
T2-3	HE-H	90928	80589	78647	72837	54716
T2-4	HE-H	87720	77559	76119	70781	49704
T2-5	HE-H	97913	86257	84582	80179	64768
T2-6	HE-H	100987	88649	87616	83072	58996

Number of non-chimeric reads were used for investigation of gut-microbiota after filtered out by DADA2.

CON; group fed with a diet without *Hericium erinaceus*, HE-L; group fed with a diet of 0.4 g *Hericium erinaceus*, HE-H; group fed with a diet of 0.8 g *Hericium erinaceus*; *n* = 6 per group.

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Table 3. Analysis of gut microbial alpha diversity in dogs after feeding diets with *Hericium erinaceus*

Alpha diversity	Group			Kruskal-Wallis
	CON	HE - L	HE - H	*p*-value
Observed features ± SD	192.43 ± 39.22	195.43 ± 51.60	180.86 ± 34.10	0.566
Shannon's index ± SD	5.64 ± 0.37	5.26 ± 0.66	5.37 ± 0.47	0.530
Evenness ± SD	0.75 ± 0.04	0.70 ± 0.07	0.72 ± 0.05	0.139
Chao1 estimates ± SD	192.60 ± 39.15	195.58 ± 51.63	181.13 ± 34.23	0.561

CON; group fed with a diet without *Hericium erinaceus*, HE-L; group fed with a diet of 0.4 g *Hericium erinaceus*, HE-H; group fed with a diet of 0.8 g *Hericium erinaceus*; n = 6 per group.

g per body weight HE (HE-L), alpha diversity of the mean values of observed features, Shannon index, Evenness, and Choa1 index were 195.4 ± 51.6, 5.2 ± 0.66, 0.7 ± 0.07, and 195.5 ± 51.63, respectively. The richness of microbial diversity in HE-L tended to increase compared with that of the CON, but there was no significant difference. In the group fed with a diet of 0.8 g HE per body weight (HE-H), the observed feature, Shannon index, Evenness, and Chao1 indices were 180.8 ± 34.10, 5.3 ± 0.47, 0.7 ± 0.05, and 181.1 ± 34.23, respectively. The richness of microbial diversity in HE-H showed a tendency to decrease compared to that of the CON, but there was no significant difference. Based on the results, the gut microbial environment of aged dogs was not disturbed by *H. erinaceus* with respect to the richness and diversity of microbiota. Alpha diversity reflects the overall degree of richness and evenness and shows the microbiome diversity within a sample. After feeding *H. erinaceus* to aged dogs, the gut microbiome diversity was unchanged and stably maintained. However, earlier reports showed different results compared to that of the current study. The richness of gut microbiota in Kunming mice was increased when mouse were gavaged for 28 days with *H. erinaceus* at 500 mg/kg and 1,000 mg/kg, respectively [26]. Furthermore, the Shannon index increased in BALB/c mice when 100 mg/kg of *H. erinaceus* was orally administrated for 21 days [33]. Despite the *H. erinaceus* concentration being higher than that used in our experiments, the species difference between mice and dogs would be affected. Additionally, Turner [34] reported that the anatomical position of the digestive system had an important effect on the evaluation of gut microbiota.

Next, the weighted and unweighted unique fraction metric (UniFrac) distance was analyzed to confirm the difference in the gut-microbiota composition changed by *H. erinaceus* intake among groups (Fig. 1). Hence, HE-H had a significant difference with the CON in weighted Unifrac distance (p = 0.047). It implied that diet intake with 0.8 g *H. erianceus* for 16 weeks changed the gut microbiota composition in aged dogs. However, there was no difference in the unweighted Unifrac distance which is consistent with no difference in the microbial richness. HE-L showed no different distance of microbiota communities in both CON and HE-H. Therefore, HE-L did not critically affect the gut microbiota ecosystems, and HE-H changed the proportion of microbes although it did not result in a significant difference in alpha diversity measurements.

Taxonomic assignment and gut microbial compositions

To determine the effect of *H. erinaceus* intake on the proportion of gut microbial, ASVs were taxonomically classified using the q2-feature-classifier (Fig. 2). At the phylum level in the CON, *Firmicutes* (52.29%), *Bacteroidetes* (28.34%), and *Fusobacteriota* (11.43%) were the most relatively abundant taxa. The top 10 relatively abundant genus were *Bacteroides* (14.54%), *Faecalibacterium* (9.57%), *Fusobacterium* (9.36%), *Blautia* (8.32%), *Peptoclostridium* (5.91%), *Prevotella* (5.48%), *Alloprevotella* (4.92%), *Escheriachia-Shigella* (3.15%), *Clostridium sensu stricto*_1 (2.83%), and *Ruminococcus gnavus* group (2.19%). *Firmicutes,* which was the most dominant microbe at the

phylum level, was reduced in both HE-L (47.99%) and HE-H (29.58%) after feeding on a diet containing *H. erinaceus* for 16 weeks. Contrastingly, *Bacteroidetes* in HE-L and HE-H increased to 28.94% and 51.19%, respectively. At the genus level, *Fusobacterium* (17.78%), *Bacteroidetes* (15.09%), and *Blautia* (12.47%) were the most relatively abundant taxa in HE-L. At the genus level in HE-H, *Bacteroidetes* (22.22%), *Prevotella* (17.08%), and *Fusobacterium* (11.66%) were the most relatively abundant taxa. *Prevotella* is known to be increased when consuming a fiber-rich diet [35], and *Blautia* was earlier reported that it is highly related to visceral fat accumulation [36]. Therefore, it would be possible that *H. erinaceus* is related to anti-obesity. The pair-end *t*-test results by the relative abundance of ASVs showed that there was no difference in genera between the CON and HE-L (data are not shown). In HE-H, when compared with CON, the genus *Prevotella* increased ($p = 0.044$), and *Tyzzerella*, a genus of the family *Lachnospiraceae*, reduced ($p = 0.027$).

LEfSe analysis was performed to identify biomarkers among groups (Fig. 3). First, the results showed that the difference between the CON and HE-L was not found at the phylum level, and

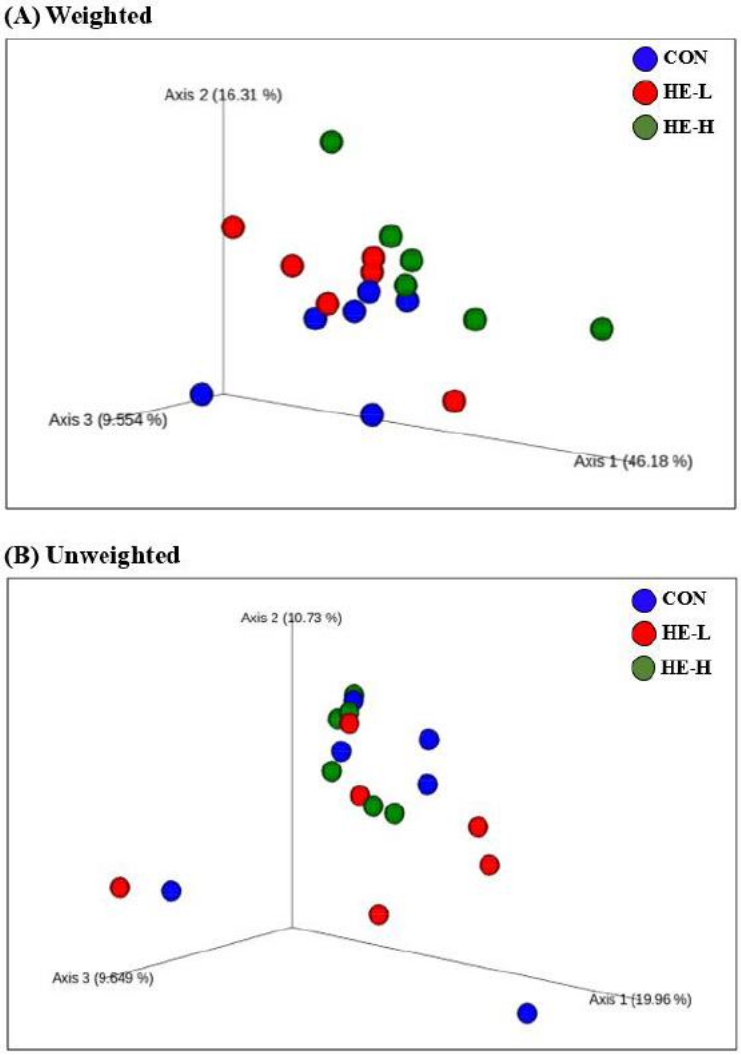

Fig. 1. Plots of the principal coordinates analysis (PCoA) of QIIME identified amplicon sequencing variants (ASVs). (A) Weighted unique fraction metric (Unifrac) distance. HE-H group clustered separately from CON ($p < 0.05$). (B) Unweighted Unifrac distance. Each of the axes were indicated component numbers of % variance between samples. CON; group fed with a diet without *Hericium erinaceus*, HE-L; group fed with a diet of 0.4 g *Hericium erinaceus*, HE-H; group fed with a diet of 0.8 g *Hericium erinaceus*; n = 6 per group.

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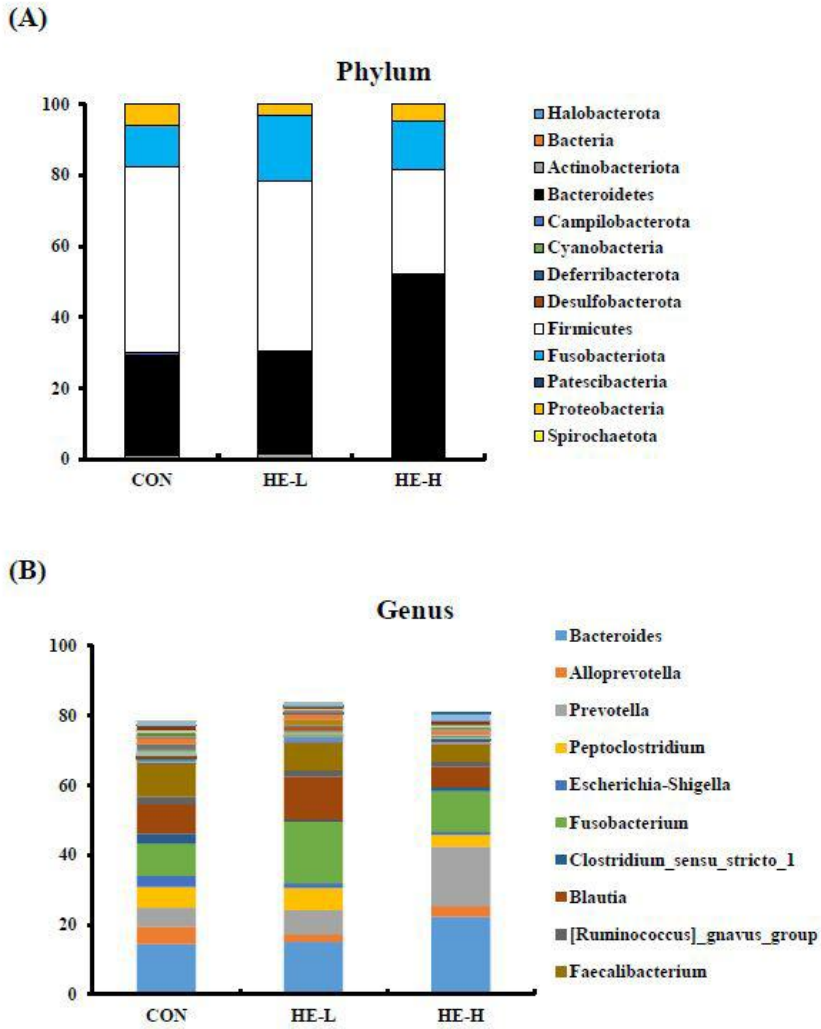

Fig. 2. Bar plots depicting the relative abundance of gut-microbiota in dog after feeding diets with *Hericium erinaceus* at (A) phylum and (B) genus levels. The Y-axis is the relative abundance of microbiota composition. Total taxonomy classification is shown at the phylum level. The graph at genus level displayed the total microbiota, but the name annotations only displayed the relative abundance of top 10 microbes. CON; group fed with a diet without *Hericium erinaceus*, HE-L; group fed with a diet of 0.4 g *Hericium erinaceus*, HE-H; group fed with a diet of 0.8 g *Hericium erinaceus*; *n* = 6 per group.

HE-L showed only a decrease in the genera *Propionigenum* (Fig. 3A). Comparing CON and HE-H, *Bacteroidetes* significantly increased in HE-H and *Firmicutes* significantly increased in CON at the phylum level (Fig. 3B). Microbes that are included in *Bacteroidetes* at the phylum, such as class *Bacteroidia* and order *Bacteroidales* also increased in HE-H (Fig. 3C). Additionally, microbes that are included in phylum *Firmicutes*, such as *Clostridia* at class, *Lachnospirales* and *Lactobacillales* at order, *Lachnospiraceae, Campylobacteraceae*, and *Streptococcaceae* at family, and *Campylobacter*, *Streptococcus*, and *Tyzzerella* at genus were found in high level in the CON group. *Firmicutes* and *Bacteroidetes* were reported to be the predominant phyla in the gut of dogs [37–39]. The ratio of Firmicutes/Bacteroidetes (F/B) was associated with the balance of the microbiome ecosystem and an appropriate ratio of F/B is important to create a healthy gut microbiota community [40]. Many studies reported that the ratio of F/B increases in obese dogs [41,42]. *Firmicutes* was a phyla that is increased by a high-calorie diet and mostly in obesity [43–45]. *Bacteroidetes* were associated with

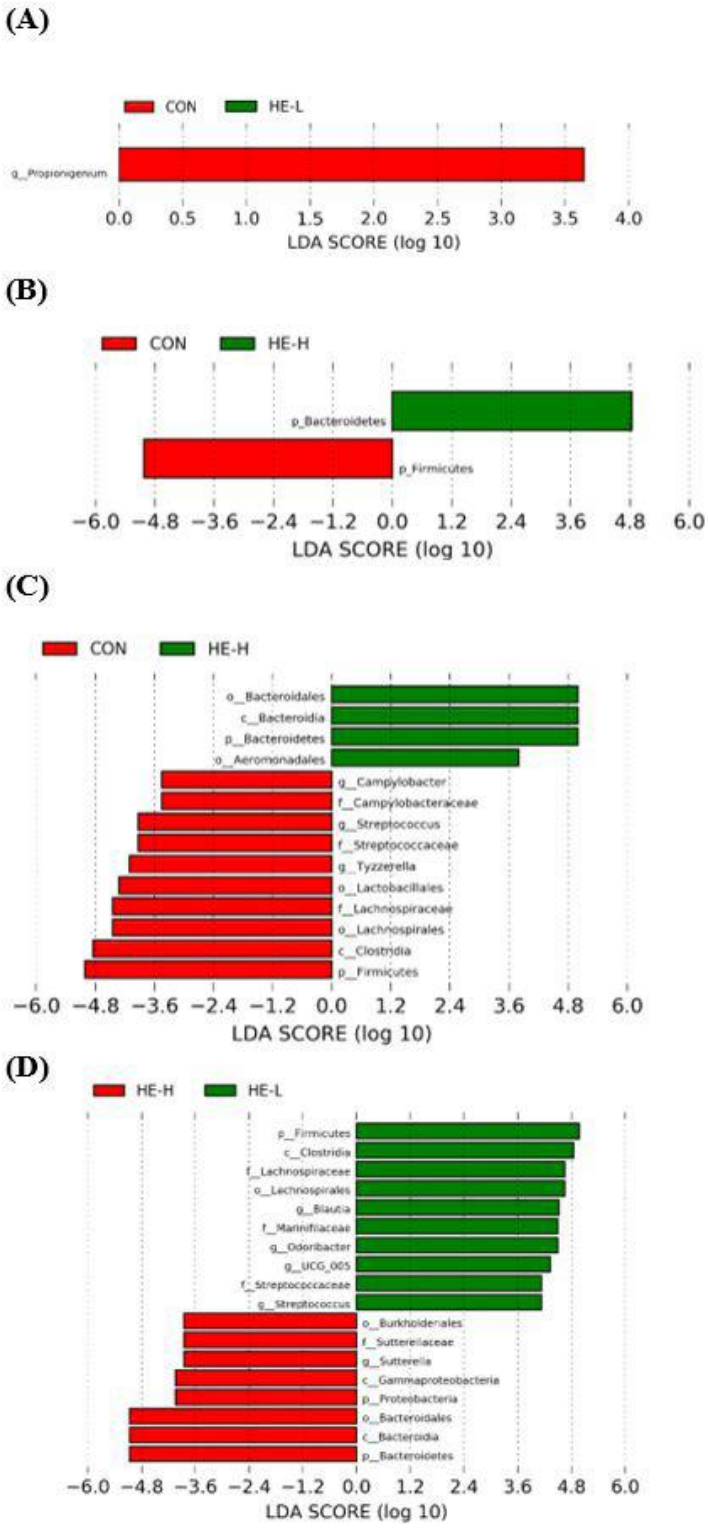

Fig. 3. Plots showcasing the linear discriminant analysis (LDA) effect size (LEfSe) bar of dog on gut microbiota between groups. (A) Gut microbiota between CON and HE-L at the genus level. (B) Gut microbiota between CON and HE-H at the phylum level. (C) Gut microbiota between CON and HE-H at the genus level. (D) Gut microbiota between HE-L and HE-H at the genus level. The bars represents the size of the difference microbes, and the two colors represent the two other groups. LDA score < 3.0, CON; group fed with a diet without *Hericium erinaceus*, HE-L; group fed with a diet of 0.4 g *Hericium erinaceus*, HE-H; group fed with a diet of 0.8 g *Hericium erinaceus*; g_, genus; p_, phylum; o_, order; c_, class; f_, family; n = 6 per group.

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weight loss and abundant phylum taxa of dogs fed a diet of high-protein and low carbohydrate [43,46]. In the current study, we found that the decrease in the F/B ratio would reflect the anti-obesity ability of *H. erinaceus*. However, there is limited information on the F/B ratio in dogs, and it is necessary to accumulate microbiome data through various studies on dogs.

Additionally, the reduction of genera *Campylobacter*, *Streptococcus*, and *Tyzzerella* in HE-H were observed. *Campylobacter* infections are one of the most common causes of diarrhea in humans, and it is a genus in which various probiotics are developed to reduce the abundance in the gut [47–49]. The abundance of *Streptococcus* was associated with inflammatory reaction and atherosclerosis [50,51]. *Tyzzerella* was increased in cardiovascular disease risk, and also asymptomatic and symptomatic in patients with viral infections severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) increased in the gut [52,53]. Additionally, *Tyzzerella* was related to chronic intestinal inflammation by change of abundance [54]. Furthermore, the order of *Aeromonadales* increased even though the family *Campylobacteraceae* and its genus *Campylobacter* which are included at the same phylum level reduced in HE-H. It could be the result of positive interaction between those microorganisms for mutual competition or survival, which is an important part of the evolution of microorganisms [55–58]. In the comparative analysis between the two concentrations (0.4 g and 0.8 g) of *H. erinaceus,* no significant difference was found (Fig. 3D). The optimal concentration of *H. erinaceus* is important in terms of the safety of pet food. In this study, we found that the concentration of *H. erinaceus* can maintain the richness and diversity of the gut microbial environment of dogs. It would support that *H. erinaceus* would be safe to be used in pet foods.

H. erinaceus is known to have health benefits including anti-inflammation and anti-obesity [59,60]. In this study, we investigated the changes in gut microbial abundance induced by diet with *H. erinaceus* in aged dogs. The population of genera *Campylobacter*, *Streptococcus,* and *Tyzzerella* is involved in inflammation, and the F/B ratio which is related to obesity was reduced after feeding on *H. erinaceus*. It would support that the intake of *H. erinaceus* improves the immunity of aged dogs and helps body weight control by regulating the gut microbial environment. It may provide insights into the possibility of *H. erinaceus* as a functional ingredient used in pet foods. As the data related to *H. erinaceus* intake and its effect on the microbiome in dogs are limited, this result can be useful information to further study the influence of *H. erinaceus* in dogs.

REFERENCES

1. Larsen JA, Farcas A. Nutrition of aging dogs. Vet Clin North Am Small Anim Pract. 2014;44:741-59. https://doi.org/10.1016/j.cvsm.2014.03.003
2. McKenzie BA, Chen FL, Gruen ME, Olby NJ. Canine geriatric syndrome: a framework for advancing research in veterinary geroscience. Front Vet Sci. 2022;9:853743. https://doi.org/10.3389/fvets.2022.853743
3. Kaeberlein M, Creevy KE, Promislow DEL. The dog aging project: translational geroscience in companion animals. Mamm Genome. 2016;27:279-88. https://doi.org/10.1007/s00335-016-9638-7
4. Hall JA, Jewell DE. Feeding healthy beagles medium-chain triglycerides, fish oil, and carnitine offsets age-related changes in serum fatty acids and carnitine metabolites. PLOS ONE. 2012;7:e49510. https://doi.org/10.1371/journal.pone.0049510
5. Akdoğan Kaymaz A. The place of geroprotective agents in life quality and longevity of companion animals. In: Rizvi S, Çakatay U, editors. Molecular basis and emerging strategies for anti-aging interventions. Singapore: Springer; 2018. p. 373-86.

6. Hou Y, Ding X, Hou W. Composition and antioxidant activity of water-soluble oligosaccharides from Hericium erinaceus. Mol Med Rep. 2015;11:3794-9. https://doi.org/10.3892/mmr.2014.3121
7. Qin T, Liu X, Luo Y, Yu R, Chen S, Zhang J, et al. Characterization of polysaccharides isolated from Hericium erinaceus and their protective effects on the DON-induced oxidative stress. Int J Biol Macromol. 2020;152:1265-73. https://doi.org/10.1016/j.ijbiomac.2019.10.223
8. Hou X, Liu J, Li Z, Chang M, Guo M, Feng C, et al. Fruiting body polysaccharides of Hericium erinaceus induce apoptosis in human colorectal cancer cells via ROS generation mediating caspase-9-dependent signaling pathways. Food Funct. 2020;11:6128-38. https://doi.org/10.1039/D0FO00916D
9. Kim YO, Lee SW, Oh CH, Rhee YH. Hericium erinaceus suppresses LPS-induced pro-inflammation gene activation in RAW264.7 macrophages. Immunopharmacol Immunotoxicol. 2012;34:504-12. https://doi.org/10.3109/08923973.2011.633527
10. Mori K, Ouchi K, Hirasawa N. The anti-inflammatory effects of lion's mane culinary-medicinal mushroom, Hericium erinaceus (higher Basidiomycetes) in a coculture system of 3T3-L1 adipocytes and RAW264 macrophages. Int J Med Mushrooms. 2015;17:609-18. https://doi.org/10.1615/IntJMedMushrooms.v17.i7.10
11. Ma BJ, Shen JW, Yu HY, Ruan Y, Wu TT, Zhao X. Hericenones and erinacines: stimulators of nerve growth factor (NGF) biosynthesis in Hericium erinaceus. Mycology. 2010;1:92-8. https://doi.org/10.1080/21501201003735556
12. Yankner BA, Shooter EM. The biology and mechanism of action of nerve growth factor. Annu Rev Biochem. 1982;51:845-68. https://doi.org/10.1146/annurev.bi.51.070182.004213
13. Pierchala BA, Ahrens RC, Paden AJ, Johnson EM Jr. Nerve growth factor promotes the survival of sympathetic neurons through the cooperative function of the protein kinase C and phosphatidylinositol 3-kinase pathways. J Biol Chem. 2004;279:27986-93. https://doi.org/10.1074/jbc.M312237200
14. Obara Y, Nakahata N. The signaling pathway of neurotrophic factor biosynthesis. Drug News Perspect. 2002;15:290-8. https://doi.org/10.1358/dnp.2002.15.5.840042
15. Qiu Y, Chen Z, Su E, Wang L, Sun L, Lei P, et al. Recent strategies for the biosynthesis of ergothioneine. J Agric Food Chem. 2021;69:13682-90. https://doi.org/10.1021/acs.jafc.1c05280
16. Roda E, Ratto D, De Luca F, Desiderio A, Ramieri M, Goppa L, et al. Searching for a longevity food, we bump into Hericium erinaceus primordium rich in ergothioneine: the "longevity vitamin" improves locomotor performances during aging. Nutrients. 2022;14:1177. https://doi.org/10.3390/nu14061177
17. Dubost NJ, Ou B, Beelman RB. Quantification of polyphenols and ergothioneine in cultivated mushrooms and correlation to total antioxidant capacity. Food Chem. 2007;105:727-35. https://doi.org/10.1016/j.foodchem.2007.01.030
18. Cheah IK, Halliwell B. Ergothioneine, recent developments. Redox Biol. 2021;42:101868. https://doi.org/10.1016/j.redox.2021.101868
19. Kim S. Antioxidant compounds for the inhibition of enzymatic browning by polyphenol oxidases in the fruiting body extract of the edible mushroom Hericium erinaceus. Foods. 2020;9:951. https://doi.org/10.3390/foods9070951
20. Inanaga K. Marked improvement of neurocognitive impairment after treatment with compounds from Hericium erinaceum: a case study of recurrent depressive disorder. Pers Med Univ. 2014;3:46-8. https://doi.org/10.1016/j.pmu.2014.02.004
21. Lee KF, Chen JH, Teng CC, Shen CH, Hsieh MC, Lu CC, et al. Protective effects of

Hericium erinaceus mycelium and its isolated erinacine A against ischemia-injury-induced neuronal cell death via the inhibition of iNOS/p38 MAPK and nitrotyrosine. Int J Mol Sci. 2014;15:15073-89. https://doi.org/10.3390/ijms150915073
22. Chang HC, Yang HL, Pan JH, Korivi M, Pan JY, Hsieh MC, et al. Hericium erinaceus inhibits TNF-α-induced angiogenesis and ROS generation through suppression of MMP-9/NF-κB signaling and activation of Nrf2-mediated antioxidant genes in human EA.hy926 endothelial cells. Oxid Med Cell Longev. 2016;2016:8257238. https://doi.org/10.1155/2016/8257238
23. Holscher HD. Dietary fiber and prebiotics and the gastrointestinal microbiota. Gut Microbes. 2017;8:172-84. https://doi.org/10.1080/19490976.2017.1290756
24. Ozdal T, Sela DA, Xiao J, Boyacioglu D, Chen F, Capanoglu E. The reciprocal interactions between polyphenols and gut microbiota and effects on bioaccessibility. Nutrients. 2016;8:78. https://doi.org/10.3390/nu8020078
25. Pérez-Reytor D, Puebla C, Karahanian E, García K. Use of short-chain fatty acids for the recovery of the intestinal epithelial barrier affected by bacterial toxins. Front Physiol. 2021;12:650313. https://doi.org/10.3389/fphys.2021.650313
26. Yang Y, Ye H, Zhao C, Ren L, Wang C, Georgiev MI, et al. Value added immunoregulatory polysaccharides of Hericium erinaceus and their effect on the gut microbiota. Carbohydr Polym. 2021;262:117668. https://doi.org/10.1016/j.carbpol.2021.117668
27. Wilmanski T, Diener C, Rappaport N, Patwardhan S, Wiedrick J, Lapidus J, et al. Gut microbiome pattern reflects healthy ageing and predicts survival in humans. Nat Metab. 2021;3:274-86. https://doi.org/10.1038/s42255-021-00377-0
28. Bolyen E, Rideout JR, Dillon MR, Bokulich NA, Abnet CC, Al-Ghalith GA, et al. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. Nat Biotechnol. 2019;37:852-7. https://doi.org/10.1038/s41587-019-0209-9
29. Martin M. Cutadapt removes adapter sequences from high-throughput sequencing reads. EMBnet J. 2011;17:10-2. https://doi.org/10.14806/ej.17.1.200
30. Callahan BJ, McMurdie PJ, Rosen MJ, Han AW, Johnson AJA, Holmes SP. DADA2: high-resolution sample inference from Illumina amplicon data. Nat Methods. 2016;13:581-3. https://doi.org/10.1038/nmeth.3869
31. Quast C, Pruesse E, Yilmaz P, Gerken J, Schweer T, Yarza P, et al. The SILVA ribosomal RNA gene database project: improved data processing and web-based tools. Nucleic Acids Res. 2013;41:D590-6. https://doi.org/10.1093/nar/gks1219
32. Paulson JN, Stine OC, Bravo HC, Pop M. Differential abundance analysis for microbial marker-gene surveys. Nat Methods. 2013;10:1200-2. https://doi.org/10.1038/nmeth.2658
33. Wang D, Zhu X, Tang X, Li H, Yizhen X, Chen D. Auxiliary antitumor effects of fungal proteins from Hericium erinaceus by target on the gut microbiota. J Food Sci. 2020;85:1872-90. https://doi.org/10.1111/1750-3841.15134
34. Turner PV. The role of the gut microbiota on animal model reproducibility. Anim Model Exp Med. 2018;1:109-15. https://doi.org/10.1002/ame2.12022
35. De Filippo C, Cavalieri D, Di Paola M, Ramazzotti M, Poullet JB, Massart S, et al. Impact of diet in shaping gut microbiota revealed by a comparative study in children from Europe and rural Africa. Proc Natl Acad Sci USA. 2010;107:14691-6. https://doi.org/10.1073/pnas.1005963107
36. Ozato N, Saito S, Yamaguchi T, Katashima M, Tokuda I, Sawada K, et al. Blautia genus associated with visceral fat accumulation in adults 20–76 years of age. NPJ Biofilms Microbiomes. 2019;5:28. https://doi.org/10.1038/s41522-019-0101-x
37. Chun JL, Ji SY, Lee SD, Lee YK, Kim B, Kim KH. Difference of gut microbiota composition

based on the body condition scores in dogs. J Anim Sci Technol. 2020;62:239-46. https://doi.org/10.5187/jast.2020.62.2.239

38. Handl S, Dowd SE, Garcia-Mazcorro JF, Steiner JM, Suchodolski JS. Massive parallel 16S rRNA gene pyrosequencing reveals highly diverse fecal bacterial and fungal communities in healthy dogs and cats. FEMS Microbiol Ecol. 2011;76:301-10. https://doi.org/10.1111/j.1574-6941.2011.01058.x
39. Hooda S, Minamoto Y, Suchodolski JS, Swanson KS. Current state of knowledge: the canine gastrointestinal microbiome. Anim Health Res Rev. 2012;13:78-88. https://doi.org/10.1017/S1466252312000059
40. Stojanov S, Berlec A, Štrukelj B. The influence of probiotics on the firmicutes/bacteroidetes ratio in the treatment of obesity and inflammatory bowel disease. Microorganisms. 2020;8:1715. https://doi.org/10.3390/microorganisms8111715
41. Ley RE, Bäckhed F, Turnbaugh P, Lozupone CA, Knight RD, Gordon JI, et al. Obesity alters gut microbial ecology. Proc Natl Acad Sci USA. 2005;102:11070-5. https://doi.org/10.1073/pnas.0504978102
42. Turnbaugh PJ, Ley RE, Mahowald MA, Magrini V, Mardis ER, Gordon JI. An obesity-associated gut microbiome with increased capacity for energy harvest. Nature. 2006;444:1027-31. https://doi.org/10.1038/nature05414
43. Million M, Lagier JC, Yahav D, Paul M. Gut bacterial microbiota and obesity. Clin Microbiol Infect. 2013;19:305-13. https://doi.org/10.1111/1469-0691.12172
44. Turnbaugh PJ, Bäckhed F, Fulton L, Gordon JI. Diet-induced obesity is linked to marked but reversible alterations in the mouse distal gut microbiome. Cell Host Microbe. 2008;3:213-23. https://doi.org/10.1016/j.chom.2008.02.015
45. Ley RE, Turnbaugh PJ, Klein S, Gordon JI. Human gut microbes associated with obesity. Nature. 2006;444:1022-3. https://doi.org/10.1038/4441022a
46. Li Q, Lauber CL, Czarnecki-Maulden G, Pan Y, Hannah SS. Effects of the dietary protein and carbohydrate ratio on gut microbiomes in dogs of different body conditions. mBio. 2017;8:e01703-16. https://doi.org/10.1128/mBio.01703-16
47. Balta I, Butucel E, Stef L, Pet I, Gradisteanu-Pircalabioru G, Chifiriuc C, et al. Anti-Campylobacter probiotics: latest mechanistic insights. Foodborne Pathog Dis. Forthcoming 2022. https://doi.org/10.1089/fpd.2022.0039
48. Moore JE, Corcoran D, Dooley JSG, Fanning S, Lucey B, Matsuda M, et al. Campylobacter. Vet Res. 2005;36:351-82. https://doi.org/10.1051/vetres:2005012
49. Śmiałek M, Kowalczyk J, Koncicki A. The use of probiotics in the reduction of Campylobacter spp. prevalence in poultry. Animals. 2021;11:1355. https://doi.org/10.3390/ani11051355
50. Sayols-Baixeras S, Dekkers KF, Hammar U, Baldanzi G, Lin YT, Ahmad S, et al. Streptococcus species abundance in the gut is linked to subclinical coronary atherosclerosis in 8,973 participants from the SCAPIS cohort. medRxiv [preprint]. 2022 [cited 2022 Jul 05]. https://doi.org/10.1101/2022.05.25.22275561
51. Brandsma E, Kloosterhuis NJ, Koster M, Dekker DC, Gijbels MJJ, van der Velden S, et al. A proinflammatory gut microbiota increases systemic inflammation and accelerates atherosclerosis. Circ Res. 2019;124:94-100. https://doi.org/10.1161/CIRCRESAHA.118.313234
52. Kelly TN, Bazzano LA, Ajami NJ, He H, Zhao J, Petrosino JF, et al. Gut microbiome associates with lifetime cardiovascular disease risk profile among bogalusa heart study participants. Circ Res. 2016;119:956-64. https://doi.org/10.1161/CIRCRESAHA.116.309219
53. Liu Y, Zhang H, Tang X, Jiang X, Yan X, Liu X, et al. Distinct metagenomic signatures in the SARS-CoV-2 infection. Front Cell Infect Microbiol. 2021;11:706970. https://doi.

org/10.3389/fcimb.2021.706970
54. Chen YJ, Wu H, Wu SD, Lu N, Wang YT, Liu HN, et al. Parasutterella, in association with irritable bowel syndrome and intestinal chronic inflammation. J Gastroenterol Hepatol. 2018;33:1844-52. https://doi.org/10.1111/jgh.14281
55. Decaestecker E, Gaba S, Raeymaekers JAM, Stoks R, Van Kerckhoven L, Ebert D, et al. Host–parasite 'Red Queen' dynamics archived in pond sediment. Nature. 2007;450:870-3. https://doi.org/10.1038/nature06291
56. Fisher RM, Henry LM, Cornwallis CK, Kiers ET, West SA. The evolution of host-symbiont dependence. Nat Commun. 2017;8:15973. https://doi.org/10.1038/ncomms15973
57. Ma B, Wang Y, Ye S, Liu S, Stirling E, Gilbert JA, et al. Earth microbial co-occurrence network reveals interconnection pattern across microbiomes. Microbiome. 2020;8:82. https://doi.org/10.1186/s40168-020-00857-2
58. Röttjers L, Faust K. From hairballs to hypotheses–biological insights from microbial networks. FEMS Microbiol Rev. 2018;42:761-80. https://doi.org/10.1093/femsre/fuy030
59. Kushairi N, Phan CW, Sabaratnam V, David P, Naidu M. Lion's mane mushroom, Hericium erinaceus (Bull.: Fr.) Pers. suppresses H2O2-induced oxidative damage and LPS-induced inflammation in HT22 hippocampal neurons and BV2 microglia. Antioxidants. 2019;8:261. https://doi.org/10.3390/antiox8080261
60. Hiraki E, Furuta S, Kuwahara R, Takemoto N, Nagata T, Akasaka T, et al. Anti-obesity activity of Yamabushitake (Hericium erinaceus) powder in ovariectomized mice, and its potentially active compounds. J Nat Med. 2017;71:482-91. https://doi.org/10.1007/s11418-017-1075-8